

KOR Structural — Managed-Service Findings & Redesign Record

Prepared for the T-Net (Tenacious Networks) review. Living document — maintained through the remediation program; present when complete.
Period covered: 2026-07-07 → 2026-07-25 (ongoing) · **Owner:** Ian Lalonde, KOR Structural

1. Executive summary

Between 2026-07-07 and 2026-07-25, KOR discovered and remediated a series of backup, virtualization, and data-integrity problems in the T-Net-managed environment — several of which sat unflagged for days to weeks, and one of which was a latent catastrophic data-loss condition. KOR self-drove the recovery and a full redesign. This record documents **(A)** what was found, **(B)** what KOR changed, and **(C)** the open items and questions for T-Net.

Tone is deliberately factual and evidence-based. Where T-Net’s position was sound, that is noted.

2. Findings — issues discovered

#	Finding	Evidence / date	Impact
F1	8-day silent backup outage. vCenter (192.168.1.9) management certificate expired ~07-07; all Veeam jobs failed for ~8 days until KOR restarted them manually on 07-15. No alert was raised by the managed monitoring the entire time.	Repo files show last good restore points 07-07 ~17:00; discovered manually 07-15 12:43	Permanent data loss exposure: FS01 files created 07-08→07-13 and deleted before 07-15 are unrecoverable (backup gap × shadow-copy purge).
F2	No effective monitoring / alerting. The outage, the later backup failures, and the snapshot condition (F7) all went undetected by the managed service.	Throughout	A managed backup service should detect a total backup failure same-day.
F3	Backups never restore-tested. No SureBackup, no documented restore test had ever been performed. Restorability was unproven.	First restore test performed by KOR 07-21	Backups of unknown validity = false assurance.
F4	No immutability, no GFS, no ransomware protection in the design.	Design review 07-16→07-23	No ransomware-proof copy; no deep point-in-time history.
F5	Reverse-incremental job configuration (3× write amplification) contributed to a disk-full grind during recovery.	Kor-VMs-New job; incident 07-15→	Slow, space-inefficient; part of the near-full repo condition.
F6	Data scrubbing never run on either Synology (Kor-NAS01 / Synology02). Backup-storage integrity had never been verified.	Storage Manager: “Never performed yet” (07-24)	Silent bit-rot in backups could surface only at restore time.
F7	FS01 orphaned snapshot + stranded backup disks. FS01 was left running on an un-consolidated snapshot delta, and both of FS01’s virtual disks (OS + 21 TB data) were left hot-added to the backup server (BK01) from an interrupted backup. vCenter surfaces a “consolidation needed” condition for exactly this — it was not caught.	Discovered/fixed 07-24→25	Latent catastrophic trap: a “Delete from disk” on BK01 would have destroyed FS01’s live 21 TB. Also caused ongoing FS01 backup failures and a growing snapshot.

#	Finding	Evidence / date	Impact
F8	~22.5 TB of dead/superseded backups left on the primary repo, running it to ~99% full and causing backup failures.	Reclaimed by KOR 07-24	Near-full repo breaks merges/synthetic fulls; part of the failure cascade.
F9	vCenter certificate lifecycle unmanaged — its expiry caused the entire incident. New cert now valid to 2028-07-14 ; renewal must be owned.	07-15	Same outage will recur in 2028 without an owner + reminder.
F10	Cost vs. service delivered. ~\$1,000+/month billed for managing ~3–5 VMs while F1–F9 sat unaddressed.	Ongoing	Value question for the relationship.

Where T-Net's position was sound (for balance)

- **NTFS over ReFS:** T-Net correctly advised keeping local repository storage NTFS rather than ReFS. Given BK01's 16 GB RAM, ReFS block-clone on a 25 TB+ repo would have been unsafe. Sound call.
- **Host layout:** production consolidated on the faster host (.10) with the older host (.16) held as a manual-failover reserve is a reasonable design under VMware Essentials (no vSphere HA).

3. Remediation & redesign delivered by KOR (07-15 → 07-25)

Recovery & platform - Recovered the outage: re-accepted the new vCenter cert, restarted all jobs. - Upgraded Veeam B&R **12.0.0.1420** → **12.3.2.4854**. - Stood up **APP01 (Server 2025) as the mount server** — deduplicated file-level restore now works.

Storage & hygiene - **Reclaimed ~22.5 TB** by retiring the dead FS01 reverse-incremental chain; verified live backups intact. - Converted the main job path to **forward-incremental + weekly synthetic full**. - **Data scrubbing** scheduled monthly + running on both Synologys (first-ever integrity pass). - Right-sized **DC01 16→8 GB** (freed failover margin); bumped **BK01 to 6 vCPU / 32 GB**.

Data protection & integrity - **Health checks** enabled on all 5 jobs (3 backups + 2 copy jobs). - **Ransomware detection:** inline entropy analysis (live), Veeam **Threat Hunter** signature engine, **File Detection + IoC**, alerts routed to KOR. - **Guest indexing + application-aware processing** (application-consistent backups, incl. the domain controller). - First-ever **restore test** and off-site copy build/seed.

The FS01 snapshot / stranded-disk repair (07-24→25) - Consolidated FS01's orphaned snapshot back to a clean single disk. - Removed **both** stranded FS01 disks from BK01 — **disarmed the 21 TB data-loss trap**. No data lost.

Target architecture (v5, 3-2-1-1-0) - Local immutable **Veeam Hardened Repository** (Linux/XFS) for fast ransomware recovery. - Off-site immutable **Object-Lock** tier + deep **GFS** retention. - **SureBackup** automated recovery verification.

Delivered since 2026-07-25 (update 2026-07-27)

- **SureBackup built and PROVEN:** Virtual Lab (Kor-Lab01, isolated network on the reserve host) + Application Group + job. First run **verified all four core VMs — DC01, FS01, APP01, RDS01 — boot and restore** (heartbeat + ping + role/SMB/RDP tests all pass). This is the first time KOR's backups have ever been *proven* restorable. Nightly auto-verification being scheduled.
- **Forward-forever incremental** locked in (weekly synthetic-full disabled — it was stacking with a CBT-reset full and overloading the Synology).
- **APP01 right-sized** (32→64 GB RAM, 4→8 vCPU, SQL max-memory capped) and its **SQL databases moved off the OS drive to a dedicated data disk** — with **zero downtime for the disk add** and no connection-string/code changes. Flagged a separate compliance item: APP01 runs **SQL Server Developer Edition in production** (dev/test-licensed only) — proper Standard licensing to be arranged (KOR-side, not a T-Net item).
- **DC01 right-sized** (16→8 GB), freeing manual-failover headroom on the busy host.
- **Immutability build sheet produced** — hardened-repo box speced (repurposed PC + 4×16 TB Exos SATA, Ubuntu/XFS); KOR to build.

Design decisions (recorded for the review)

- **Local VM replication to the reserve host: deferred.** For KOR’s size, manual failover + the now-*verified* Instant VM Recovery gives an acceptable RTO, and the reserve host’s **1 GbE / off-storage-fabric** path (a network-doc finding) would make on-host replicas degraded anyway.
- **Second domain controller: deferred.** SureBackup having proven DC01 restorable reduces the urgency of the DC01 single-point-of-failure; a 2nd DC remains the recommended long-term fix.

4. Open items & dependencies on T-Net

Item	Ask
Cloud Connect immutability	Enable immutability (or Insider Protection) on the KOR tenant repo — requested 07-24.
WAN accelerator	Provider-side accelerator for the off-site copy — requested.
License coverage	Confirm the rental Enterprise Plus licence covers the Hardened Repository role at no extra cost.
vCenter cert renewal	Establish ownership + reminder for the 2028-07-14 expiry (root cause of F1).
Monitoring / alerting SLA	Establish real alerting that would catch a total backup failure, a “consolidation needed” condition, and repo-full — none of which were caught (F1, F2, F7, F8).

Questions for the review

1. Why was the **8-day total backup outage** (F1) never alerted?
2. Why was the **FS01 “consolidation needed”** condition (F7) — visible in vCenter — never flagged?
3. Why were the **repeated FS01 backup failures** (F7/F8) not caught?
4. What monitoring is actually in place, and what is the alerting SLA?
5. Who owns the vCenter certificate lifecycle going forward?